Noise Robustness Test: Results and Caveats

Nov 8, 2026 · @Dor

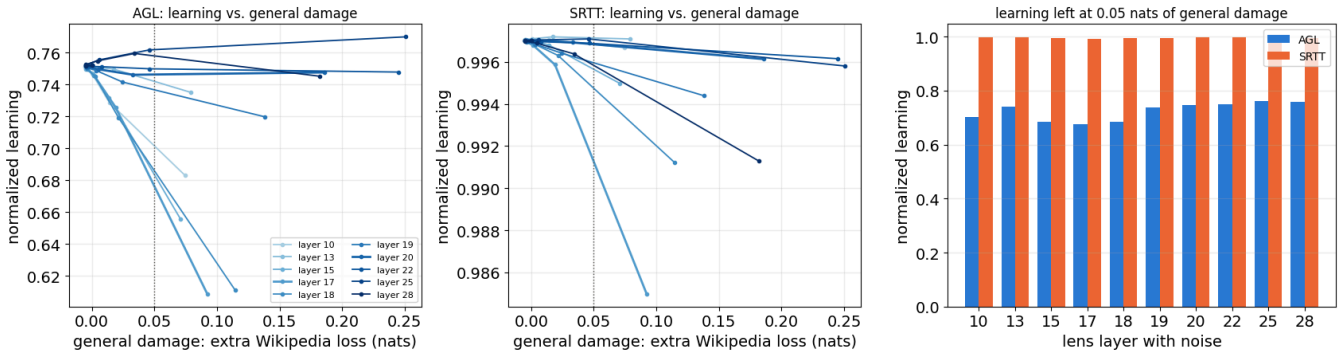
A third method agrees with the other two: at equal general damage, noise just before the induction heads (lens layers 15–18) costs about 10% of AGL learning, while noise after lens layer 19 costs none, even where it hurts general language ability most. The effect is modest, and SRTT is too robust to be affected, so this is the weakest of the three lines of evidence.

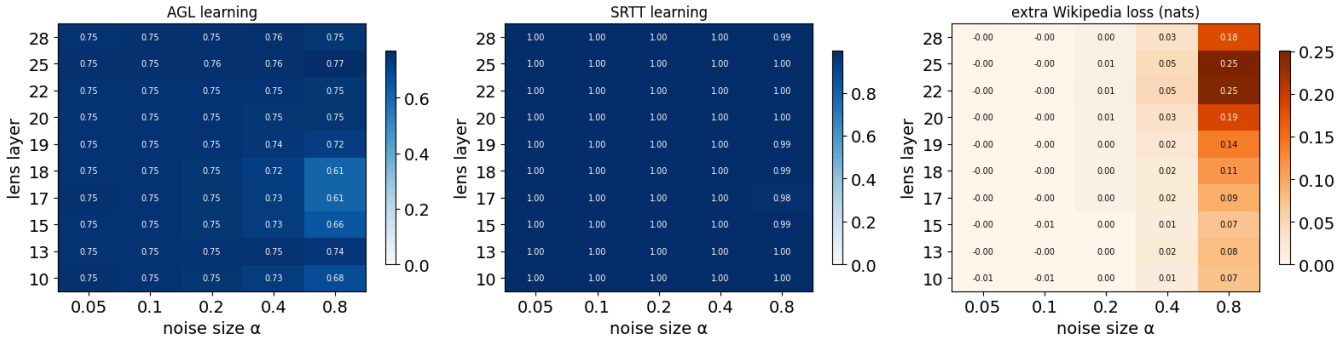
Design

At one lens layer, every word's internal state gets a random push of relative size α (0.05 to 0.8 of that state's own size). 10 layers × 5 noise sizes, 10 paired runs per grammar, plus a general-damage check on 25 Wikipedia texts.

- **Noise scaled to each word's own state size.** That makes layers comparable, even though the states grow a lot with depth.
- **BOS left alone,** with a sanity check confirming the noise has exactly the intended size and the same run always gets the same noise.
- **The main plot compares layers at equal general damage** (extra Wikipedia loss). That is the right test: without it, "layer X is fragile" could just mean "layer X matters for everything".

Results





1. For AGL, the most fragile point is just before the jump. At the same general damage (0.05 nats):

Lens layer with noise	10	13	15	17	18	**19**	20–28
AGL learning left (no noise: 0.75)	0.70	0.74	0.68	0.68	0.69	**0.74**	0.75–0.76

Noise on the state that the block-18 induction heads *read* (lens layers 15–18) costs about 10% of the learning. Noise right after them (lens layer 19 onward) costs nothing: once block 18 has done its work, disturbing the state no longer hurts the learning.

2. Late layers dissociate learning from general ability. Strong noise at layers 22–25 hurts Wikipedia prediction the most (+0.25 nats at α = 0.8), yet AGL learning does not drop at all. The in-context learning is already settled there, even though those layers still matter for ordinary text.

3. A possible echo at the earlier cluster. Layer 10 is somewhat fragile (0.70), while layer 13, right after the block-12 heads, is not (0.74). Same "before vs. after" pattern, but weaker and based on interpolation, so a hint only.

4. SRTT is almost untouchable (worst case 0.985 vs. 0.997). A sequence repeated 320 times is too redundant for noise to break, so SRTT says nothing here, matching the head ablations, where SRTT was also the more robust grammar.

Caveats

- **The effects are small.** Even the strongest noise (α = 0.8) removes only ~19% of AGL learning. Random noise in 4,096 dimensions mostly points in directions the model does not use, which is why the head ablation was the much stronger test.
- **The error bars do not test the right thing.** The ±0.016 is the spread of the *level* across 10 runs, not of the *difference* from baseline. Every run is paired (same stream with and without noise), so the per-run difference and its standard error would be much

tighter and would show whether the 0.75 → 0.68 drop is significant. The damage-matched bar plot has no error bars yet.

- **The damage-matched values are interpolated** between coarse noise sizes; for some layers 0.05 nats is only reached between α = 0.4 and 0.8, so those values are rough.
- **The noise hits every position at once:** the current word and the earlier words the heads look back at. It cannot separate "the heads cannot find the earlier word" from "the current state is confused". Sharper follow-up: noise only on earlier positions, or only on the current one.
- **The learning-curve comparison uses α = 0.4,** where the curves barely differ; α = 0.8 at layers 17 vs. 20 would show the contrast more clearly.
- **Small noise very slightly improves Wikipedia loss** (−0.005 nats at every layer). Tiny, consistent and probably harmless.